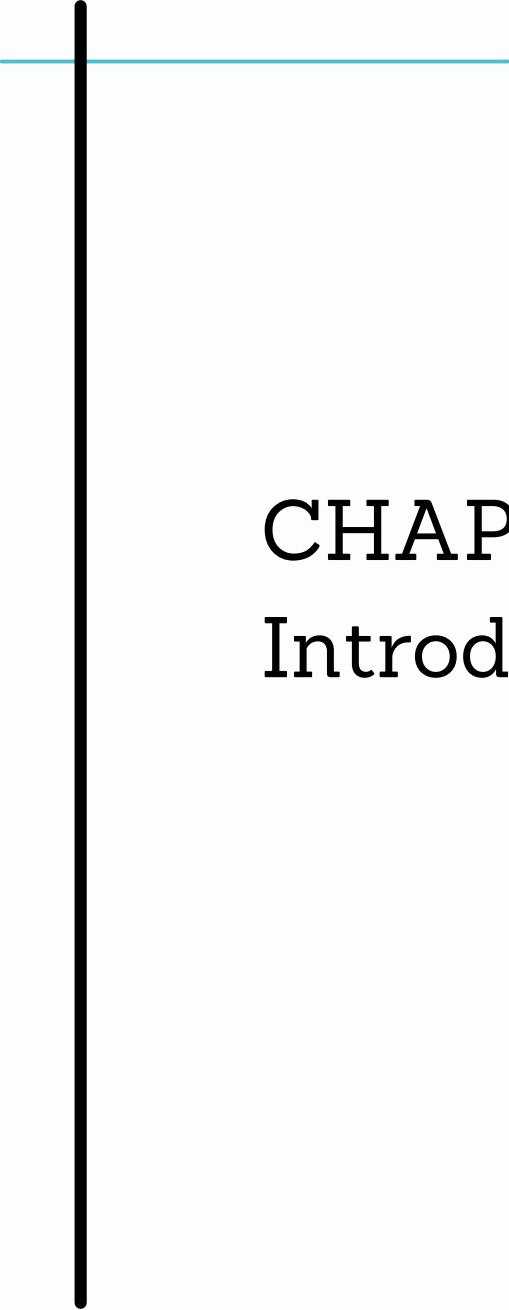




CHAPTER-1

Introduction to Artificial Intelligence

Afna K A^[L]_[SEP]



Definition

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence.

These tasks includes:

- Learning
- Problem solving
- Decision making
- Speech recognition
- Image recognition
- Language understanding

Artificial mean: Made by humans.

Intelligence means: Ability to think ,learn ,solve problems and make decisions

Examples:

- ChatGPT
- Google Maps
- Siri
- Alexa

Why Do We Need AI?

Humans have limitations.

AI helps us by:

- Speed : Humans take time to perform calculations. Processes millions of calculations quickly.
- Accuracy : Humans can make errors due to tiredness or lack of concentration. Reduces human errors.
- Automation: Performs repetitive tasks automatically.
- 24*7 Availability: Works without breaks or sleep.
- Big Data Handling: Analyzes huge amounts of information.
- Better Decision Making: Uses data to make predictions

Real-Life Applications of AI:

- AI tutors
- Google Maps
- Self-driving cars
- Face Unlock
- Disease detection

Features of AI

- **Learning** : Learns from data and improve performance
- **Reasoning**: Make Logical decisions
- **Problem Solving** : Finds the best solution
- **Decision making**: Chooses the best action using data.
- **Natural language processing**: Understands and generates human language.
- **Computer Vision** : Recognizes images and videos.
- **Automation**: Performs tasks automatically.
- **Adaptability**: Improves with experience and new data.

AI History And Evolution

1. Birth of AI (1940s–1950s):
 - Idea of intelligent machines began.
 - Alan Turing proposed that machines could think.
 - Dartmouth Conference (1956) officially introduced the term "Artificial Intelligence."
2. Early Years (1950s–1960s):
 - Rule-based AI systems.
 - AI solved simple logical problems.
3. AI Winter (1970s–1980s):
 - AI research slowed down.
 - Computers were not powerful enough.
 - Funding for AI projects decreased.
4. Rise of Expert Systems (1980s):
 - Expert systems became popular.
 - Used knowledge and rules from experts.
 - Applied in medicine, finance, and engineering.

5. Return of Neural Networks (Late 1980s–1990s):
 - Neural networks became popular again.
 - Inspired by the human brain.
 - Better learning from data.
6. Machine Learning Renaissance (2000s–Present):
 - Growth of Machine Learning.
 - Big Data became available.
 - Faster computers improved AI.
7. AI in the Modern Era (2010s–Present):
 - AI became part of daily life.
 - Recommendation systems.
 - Self-driving cars.
8. Future of AI (2020s and Beyond):
 - AI continues to improve every year.

Types of AI

Narrow AI : This is the AI we use today. It performs one specific task.

Examples:

- ChatGPT
- Google Maps
- Face Unlock
- Alexa

A chess AI can play chess extremely well but cannot cook food or drive a car

General AI: General AI would have human-level intelligence.

It could:

- Learn anything
- Solve different kinds of problems
- Understand emotions and context
- Perform many tasks like humans

Super AI: Super AI is a hypothetical form of AI that would be smarter than humans in every field.

It could outperform humans in:

- Science
- Medicine
- Creativity
- Problem solving

Advantages and Disadvantages

Advantages	Disadvantages
High Speed – AI processes large amounts of data much faster than humans.	Job Displacement – AI can automate tasks, reducing the need for some human jobs.
High Accuracy – AI provides accurate results when trained with quality data.	Security Risks – AI can be misused for cyberattacks, deepfakes, or fraud..
Handles Big Data – AI can efficiently process and analyze massive datasets.	Dependence on Data-AI learns from data.
Reduces Human Effort – AI automates repetitive and time-consuming tasks.	Cannot Think Independently: AI follows patterns learned from data.

CHAPTER-2

Understanding Artificial Intelligence



Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed for every task.

Types of ML:

- Supervised Learning
 - The computer learns using labeled data., Labeled data means both input and correct output are already known.

Fruit Color	Fruit
Red	Apple
Yellow	Banana
Orange	Orange

- Unsupervised Learning
 - The computer receives unlabeled data. There are no correct answers.The computer finds hidden patterns by itself.
 - Uses unlabeled data
 - Finds patterns
 - Groups similar data
 - Used for clustering
-

- Reinforcement Learning
 - Learn by trial and error
 - Uses rewards and penalties
 - Improves continuously
 - Used in robots and games
- Semi-supervised Learning
 - Combination of supervised and unsupervised learning
 - Uses few labeled and many unlabeled data
 - Reduces labeling effort

Deep Learning

Deep Learning is a subset of Machine Learning . It uses Artificial Neural Networks with multiple layers to learn very complex patterns. Machine Learning learns from data, but Deep Learning can learn much more complex patterns.

Neural Network

A Neural Network is a computing model inspired by the human brain

- It consists of three main layers:
 - Input Layer – Receives data.
 - Hidden Layer(s) – Processes and learns patterns
 - Output Layer – Produces the final prediction.

Example

- For face recognition:
 - Input: Image pixels
 - Hidden Layers: Learn facial features (eyes, nose, mouth, etc.)
 - Output: Person recognized or not

Data And Dataset

Data is a collection of raw facts and information.

- Raw facts
- Can be numbers, text, images, audio, or video
- Used to train AI models

Dataset

A Dataset is an organized collection of related data used for training and testing AI models

Name	Age	Marks
Rahul	20	85
Anu	21	90
Meera	19	78

Feature Engineering

A feature is an individual property or characteristic of the data.

Feature Engineering is the process of selecting, modifying, or creating useful features from raw data to improve the performance of a Machine Learning model.

Model Evaluation

Model Evaluation is the process of checking how well a Machine Learning model performs after it has been trained.

- Accuracy – How many predictions are correct.
- Precision – Out of all predicted positives, how many are actually correct?
- Recall – Out of all actual positives, how many did the model find?
- Confusion Matrix – Shows correct and incorrect predictions in detail.

Underfitting

- Underfitting occurs when a Machine Learning model is too simple and cannot learn the important patterns from the training data.
- As a result, it performs poorly on both the training data and new (test) data.

Overfitting

- Overfitting occurs when a Machine Learning model learns the training data too well, including noise and unnecessary details.
- As a result, it performs very well on training data but poorly on new (test) data.

Optimization

- Optimization is the process of finding the best values for a model's parameters (weights and biases) so that the model makes the most accurate predictions by minimizing errors (loss).
- Reduces prediction errors.
- Finds the best model parameters.
- Uses algorithms like Gradient Descent.

Regularization

- Regularization prevents a model from learning unnecessary details or noise from the training data.
- Prevents overfitting.

Bias

Bias is when an AI system gives unfair or inaccurate results because it has learned from biased or unbalanced data

- Unfair predictions

Fairness

Fairness means an AI system treats everyone equally, without discrimination based on gender, age, race, religion, or other personal characteristics.

- Equal treatment for everyone

Natural Language Processing (NLP)

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, process, and generate human language, such as English, Malayalam, Hindi, or any other language.

- ChatGPT.
- Alexa.
- Siri

Computer Vision

Computer Vision is a branch of Artificial Intelligence that enables computers to see, analyze, and understand images and videos.

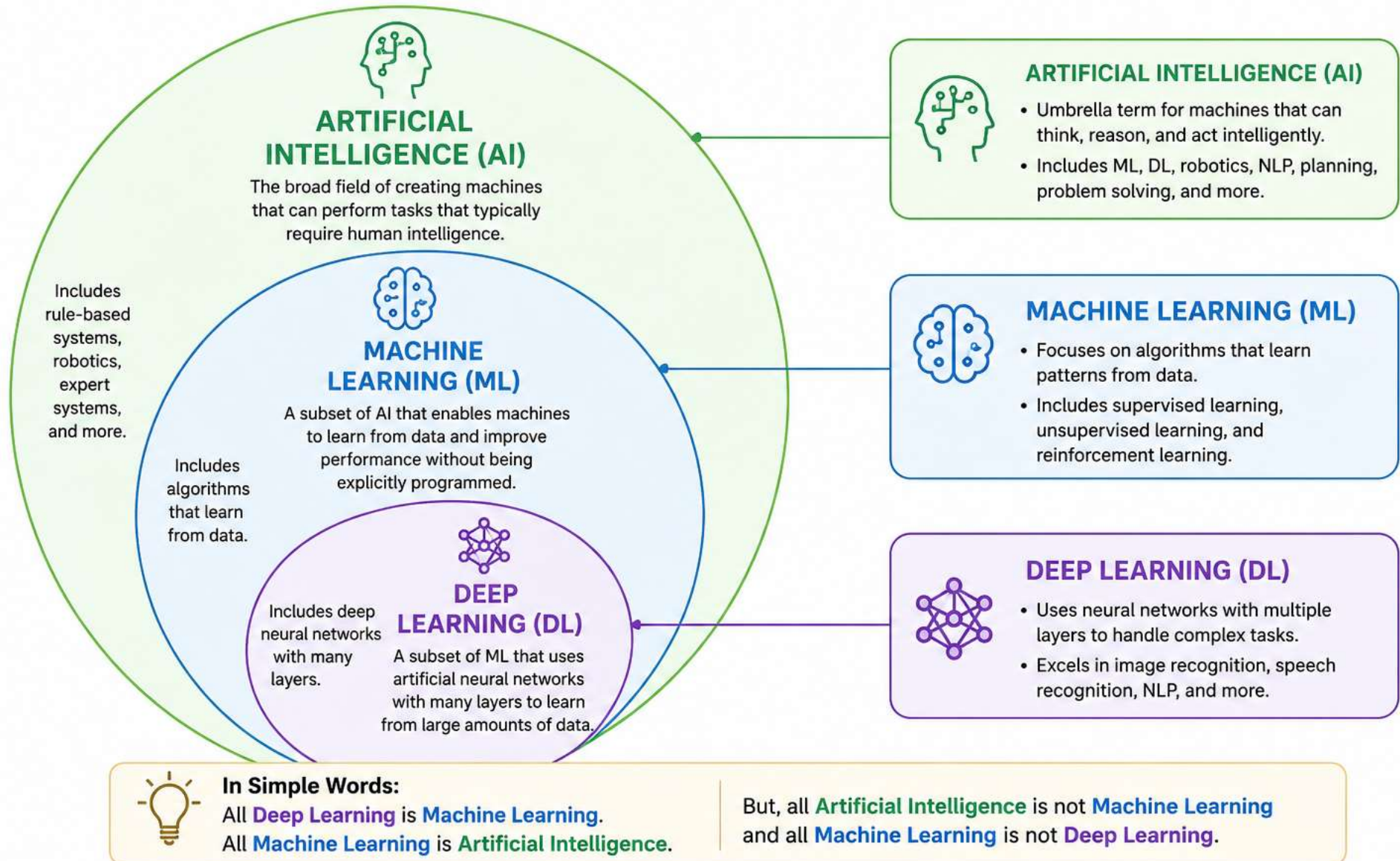
- Face Unlock
- CCTV Surveillance
- Self-driving Cars

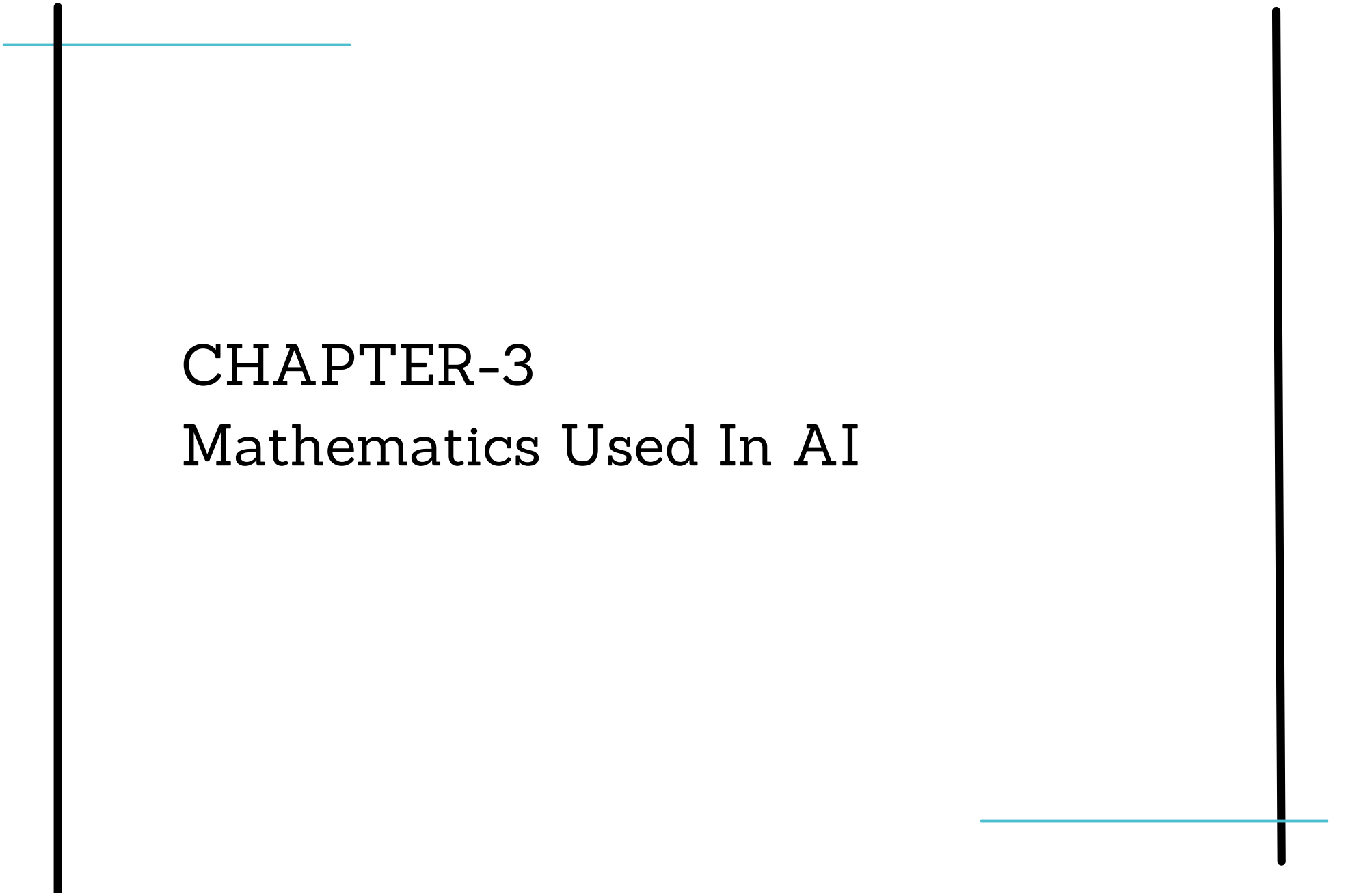
Reinforcement Learning (RL)

Reinforcement Learning is a type of Machine Learning in which an agent learns by interacting with its environment through trial and error, receiving rewards for correct actions and penalties for incorrect actions.

- Reinforcement Learning has four main components:
 - Agent
 - ❖ The learner or decision-maker.
 - ❖ Robot
 - ❖ Self-driving car
 - Environment
 - ❖ The place where the agent works.
 - ❖ Road
 - ❖ Chess board
 - Action
 - ❖ The decision taken by the agent. Alexa.
 - ❖ Turn left
 - ❖ Move forward
 - ❖ Pick up an object
 - Reward
 - ❖ Feedback from the environment.
 - ❖ Correct action → Positive reward
 - ❖ Wrong action → Negative reward (penalty)
 - The agent keeps learning until it discovers the best actions.

RELATIONSHIP BETWEEN AI, MACHINE LEARNING & DEEP LEARNING





CHAPTER-3

Mathematics Used In AI

Mathematics Used in AI

Artificial Intelligence relies on mathematics to represent data, analyze patterns, make predictions, and improve model performance. Mathematical concepts help AI systems learn from data, make decisions, and solve real-world problems efficiently.

Linear Algebra:

- Linear Algebra is the branch of mathematics that deals with vectors, matrices, and linear equations. It is used to represent, organize, and process data efficiently. Almost every Machine Learning and Deep Learning algorithm uses Linear Algebra for mathematical calculations.

Application:

- Image Processing
- Face Recognition
- Neural Networks
- Deep Learning

Vector: It is an ordered list of numbers [20, 30, 1,]

Matrix: Two dimensional arrangement of numbers in rows and columns

Calculus

Calculus is the branch of mathematics that studies change and motion. It helps AI models improve their predictions by minimizing errors during training. The two important concepts are Differentiation and Integration.

Applications:

- Gradient Descent
- Neural Networks
- Deep Learning

Probability and Statistics

Probability measures the chance of an event occurring, while Statistics helps summarize, analyze, and understand data. AI uses these concepts to make predictions even when information is uncertain.

- Descriptive statistic: Techniques For summarizing and visualizing data, includes Mean,Median,Standard deviation
- Inferential Statistics: To Test hypotheses, Evaluate probabilities and Draw conclusion about wide population

Optimization:

Optimization is the process of improving an AI model by reducing prediction errors and finding the best values for its parameters. The most commonly used optimization algorithm is Gradient Descent.

Gradient Descent: it is an iterative optimization algorithm used to minimize a cost function by adjusting model parameters in the opposite direction

Information Theory

Information Theory is the study of information and uncertainty. It helps AI measure how much useful information is available in data. One important concept is Entropy, which measures uncertainty.

Set Theory:

Set Theory is the study of collections of objects called sets. It helps AI organize data, compare groups, and identify relationships between different data elements.

Graph Theory

Graph Theory represents relationships using nodes (vertices) and edges (connections). It is widely used in AI to model networks and relationships.

Numerical Methods:

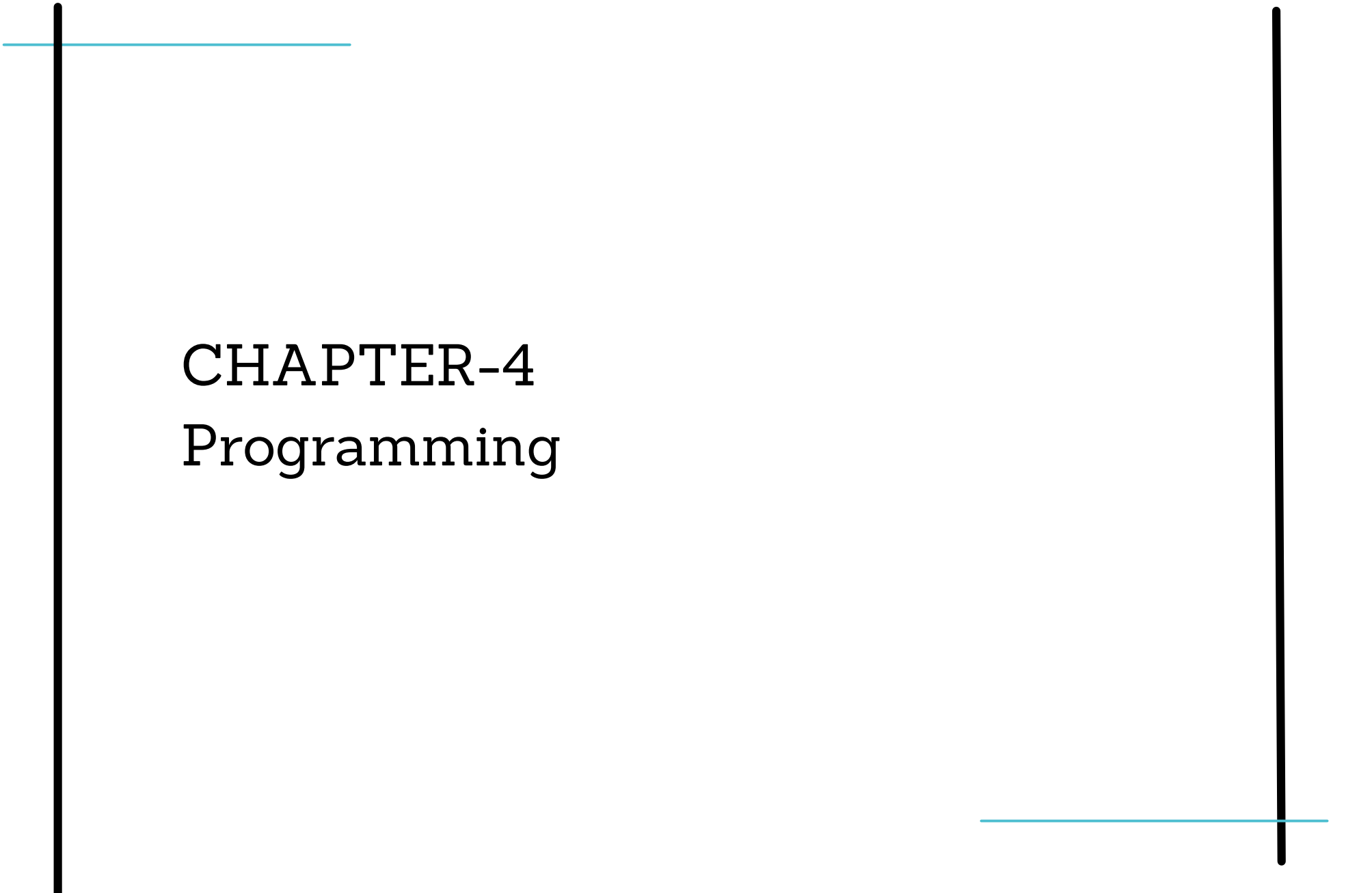
Numerical Methods are mathematical techniques used to find approximate solutions when exact solutions are difficult or impossible. They help AI solve complex equations efficiently.

- Social Media
- Google Maps

Linear Regression:

Linear Regression is a Machine Learning algorithm used to predict continuous numerical values by finding the relationship between input variables and output values.

eg: Suppose students study for different numbers of hours. Generally, students who study more tend to score higher marks. AI learns this relationship and predicts the marks of a new student. This prediction method is called Linear Regression."



CHAPTER-4

Programming

Programming in Artificial Intelligence

Programming is the process of writing instructions that tell a computer what to do. Artificial Intelligence systems are built using programming languages because they help computers process data, learn from experience, make decisions, and solve problems automatically. Programming is the tool used to build AI.

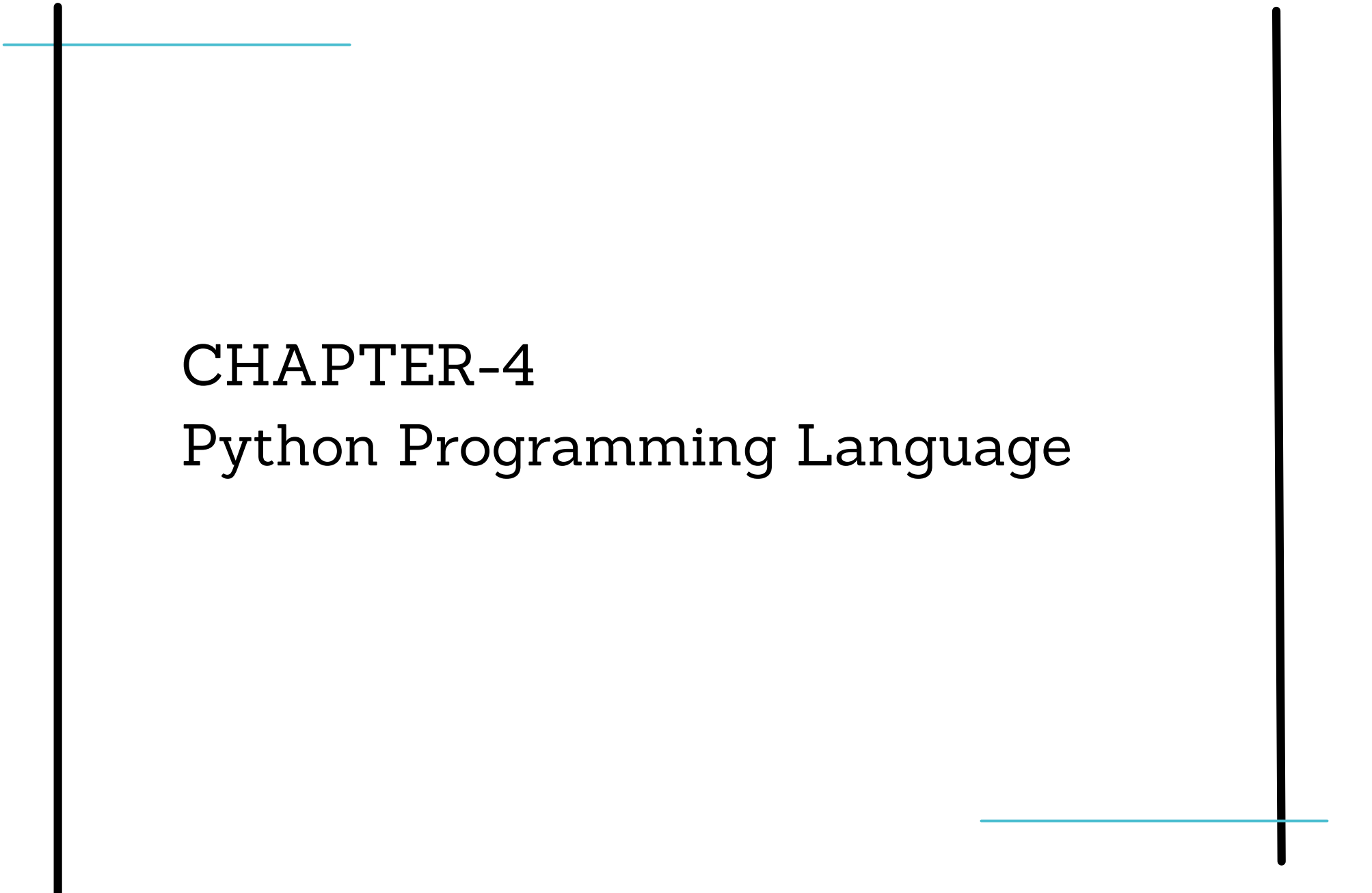
Without programming languages:

- AI cannot read data.
- AI cannot train models.
- Decision making
- AI cannot make predictions.

Programming Language	Purpose	Key Features	AI Applications
Python	The most popular language for AI and Machine Learning because it is simple and has many AI libraries.	Easy syntax, huge community, many AI libraries like NumPy, Pandas, TensorFlow, PyTorch, Scikit-learn.	Machine Learning, Deep Learning, NLP, Computer Vision, Chatbots, Data Science.
R	A programming language mainly designed for statistical computing and data analysis.	Excellent statistical functions, rich visualization packages.	Data Analysis, Statistical Modeling, Data Visualization, Research.
Java	Used to develop large and scalable AI applications.	Object-Oriented, secure, platform-independent, multithreading support.	Enterprise AI, Big Data, Search Engines, Recommendation Systems.
C++	Used when high performance and real-time execution are required.	Fast execution, direct memory management, efficient hardware control.	Robotics, Game AI, Autonomous Vehicles, Embedded AI Systems.

Julia	A modern language designed for scientific computing and AI.	High execution speed, easy mathematical programming.	Machine Learning, Scientific Computing, Numerical Analysis.
Lisp	One of the earliest AI programming languages.	Supports symbolic processing and recursion.	Expert Systems, Symbolic AI, Research.
Prolog	Logic programming language based on rules and facts.	Rule-based programming, logical inference.	Expert Systems, Knowledge Representation, Reasoning Systems.
JavaScript	Used to build AI applications that run in web browsers.	Runs directly in browsers, interactive user interfaces.	AI Chatbots, Smart Websites, Browser-based ML, Web AI.
SQL	Database language used to store and retrieve AI data.	Data querying, filtering, updating, database management.	Data Storage, Data Retrieval, Data Preprocessing.
Shell Scripting (Bash)	Automates repetitive AI development tasks.	Automation, scheduling, pipeline management.	Running AI models, Automating training, Managing datasets.

MATLAB / Octave	Used for mathematical modeling and algorithm development.	Powerful mathematical functions, visualization tools.	Image Processing, Signal Processing, Machine Learning Research.
CUDA (GPU Programming)	Enables AI models to run on NVIDIA GPUs for faster training.	Parallel computing, GPU acceleration.	Deep Learning, Large Neural Networks, High-performance Computing.



CHAPTER-4

Python Programming Language

Programming in Artificial Intelligence

Programming is the process of writing instructions that tell a computer what to do. Artificial Intelligence systems are built using programming languages because they help computers process data, learn from experience, make decisions, and solve problems automatically. Programming is the tool used to build AI.

Without programming languages:

- AI cannot read data.
- AI cannot train models.
- Decision making
- AI cannot make predictions.

Programming in Artificial Intelligence

Programming is the process of writing instructions that tell a computer what to do. Artificial Intelligence systems are built using programming languages because they help computers process data, learn from experience, make decisions, and solve problems automatically. Programming is the tool used to build AI.

Without programming languages:

- AI cannot read data.
- AI cannot train models.
- Decision making
- AI cannot make predictions.

Python is one of the most widely used programming languages in the world. It is easy to learn, simple to write, and provides many powerful libraries for Artificial Intelligence, Machine Learning, Data Science, and Deep Learning.

- **Extensive Libraries and Frameworks** :A library is a collection of pre-written code that helps programmers perform tasks easily. Python has many libraries for Artificial Intelligence.
- **Simple and Readable Syntax** : Python uses simple English-like commands. Programs written in Python are easy to read and understand. Because of its simple syntax, beginners can learn programming quickly. This is one of the main reasons why Python is widely used in AI.
- **Community Support**: Python has millions of users around the world. If programmers face any problem, they can easily find solutions through online tutorials, documentation, forums, and communities.
- **Versatility**: Versatility means the ability to perform many different tasks. Python is not only used for Artificial Intelligence.
- **Machine Learning and Deep Learning**: Python provides powerful libraries such as:
 - Scikit-learn – Machine Learning
 - TensorFlow – Deep Learning
 - PyTorch – Deep Learning
 - These libraries help developers build intelligent systems without writing complex algorithms from scratch.
- **Data Science Ecosystem** : Python widely used in data science which is closely related to AI,Many project involves data analysis, python's data Science ecosystem including jupyter notebook
- **Cross Platform Compatibility** : Python programs can run on different operating systems The same Python code usually works on all these platforms with very few changes. This makes Python highly flexible.
- **Rapid Prototyping**: Rapid prototyping means creating a working model quickly. Python allows developers to build and test AI ideas in a short time.

AI Libraries

Library	Full Form	Purpose	Key Features	Real-world Applications	📖
NumPy	Numerical Python	Numerical computing and array operations	Multi-dimensional arrays, matrices, mathematical functions, fast calculations	Data preprocessing, scientific computing, machine learning	
TensorFlow	Developed by Google	Build and train deep learning models	High performance, scalable, supports GPU, works with Keras	Image recognition, speech recognition, chatbots, self-driving cars	
PyTorch	Developed by Meta (Facebook AI Research)	Deep learning and neural networks	Dynamic computation graph, easy debugging, research-friendly	Research, NLP, computer vision, ChatGPT-like models	
Keras	High-level Deep Learning API	Easy creation of neural networks	Beginner-friendly, simple API, runs on TensorFlow	Image classification, sentiment analysis deep learning projects	
Scikit-learn	Machine Learning Library	Traditional machine learning algorithms	Classification, regression, clustering, preprocessing, model evaluation	Spam detection, recommendation systems, customer prediction	
NLTK	Natural Language Toolkit	Natural Language Processing (NLP)	Tokenization, stemming, lemmatization, POS tagging, sentiment analysis	Chatbots, language translation, text analysis	
Gensim	Topic Modeling Library	Text processing and document similarity	Word embeddings, topic modeling, document summarization	Search engines, document analysis, recommendation systems	
OpenCV	Open Source Computer Vision Library	Image and video processing	Face detection, object detection, image filtering, video analysis	Face recognition, CCTV surveillance, autonomous vehicles	

AI Libraries

Library	Purpose
NumPy	Numerical calculations
Pandas	data analysis and data manipulation.
Matplotlib	data visualization
Scikit-learn	Ready-made machine learning algorithms.
TensorFlow	building Deep Learning models.
Keras	building neural networks.
OpenCV(Open Source Computer Vision Library.)	helps computers understand images and videos.